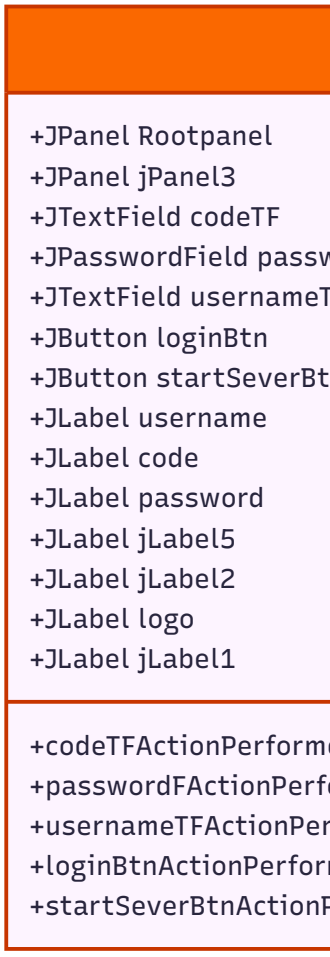
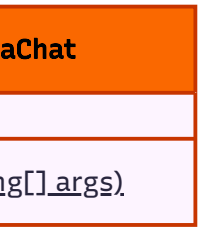
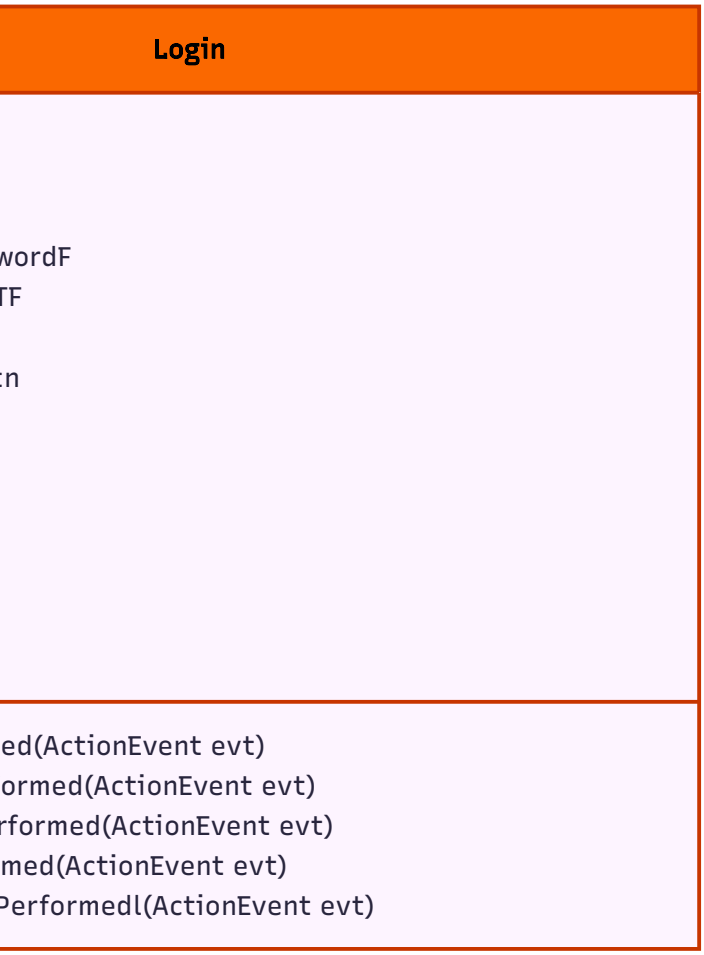
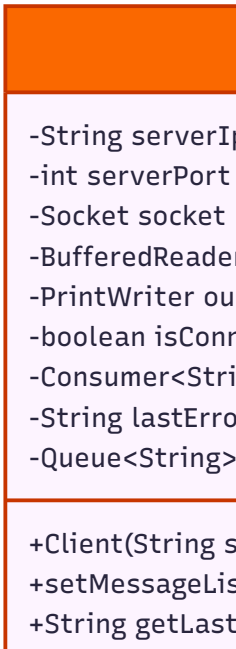

Intra
+main(Strin

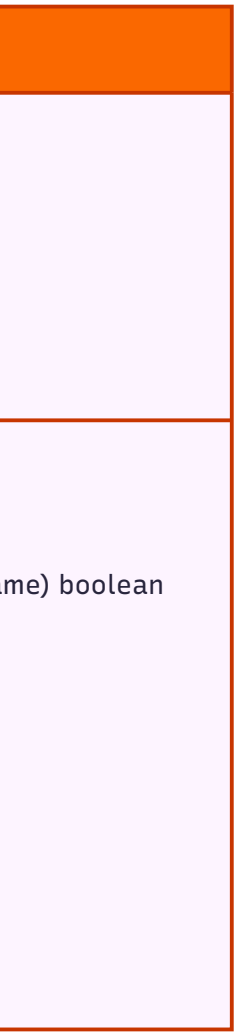

+JPanel Rootpanel
+JPanel jPanel3
+JTextField codeTF
+JPasswordField passw
+JTextField usernameT
+JButton loginBtn
+JButton startSeverBt
+JLabel username
+JLabel code
+JLabel password
+JLabel jLabel5
+JLabel jLabel2
+JLabel logo
+JLabel jLabel1
+codeTFActionPerforme
+passwordFActionPerfo
+usernameTFActionPer
+loginBtnActionPerfor
+startSeverBtnActionP

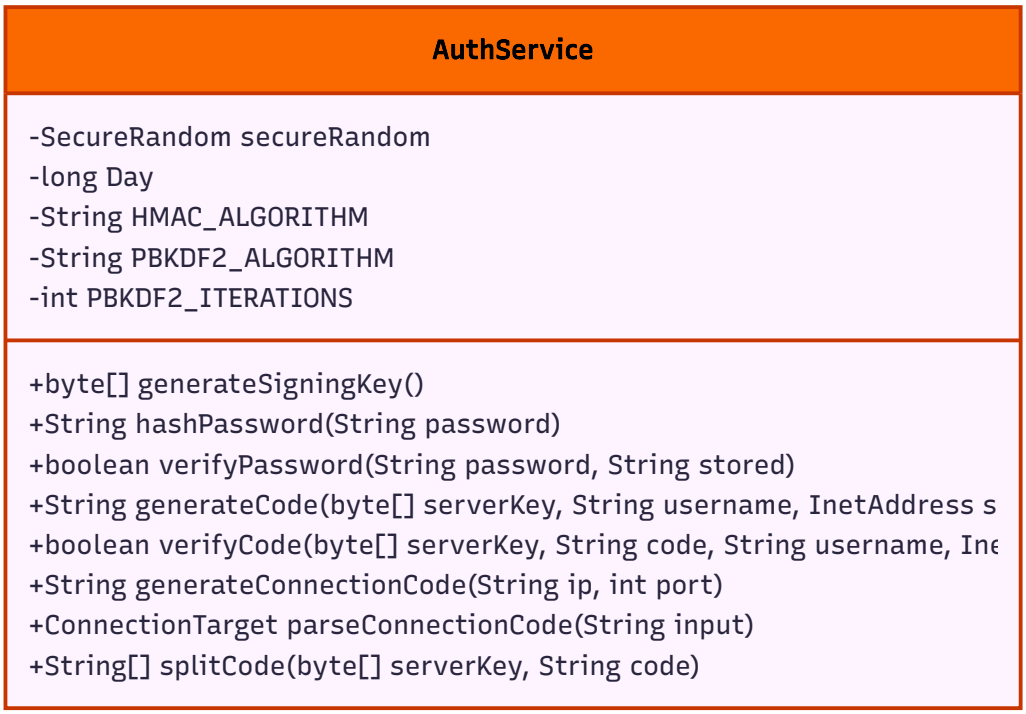

aChat
ng[] args)

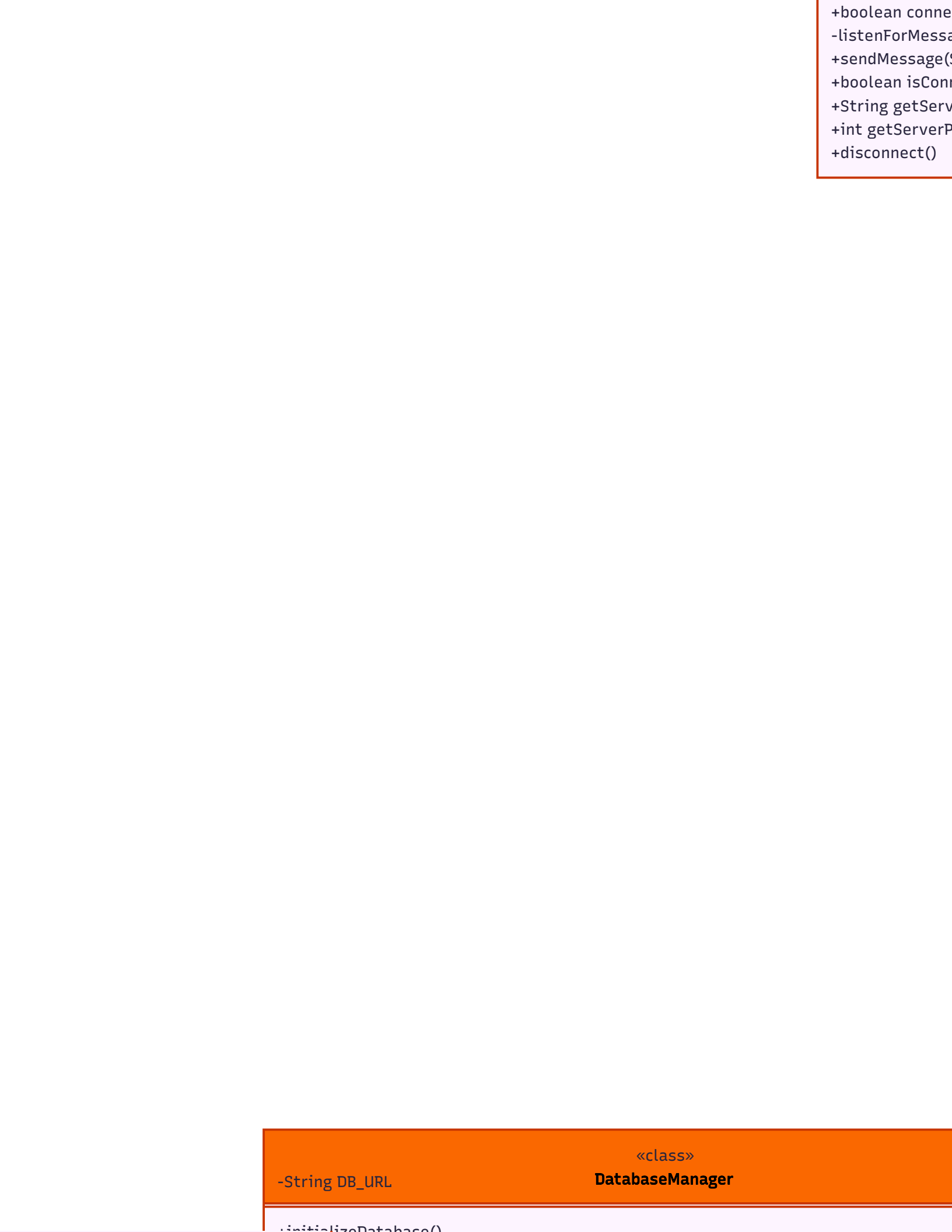

Login
wordF
TF
tn
ed(ActionEvent evt)
ormed(ActionEvent evt)
rformed(ActionEvent evt)
med(ActionEvent evt)
Performedl(ActionEvent evt)

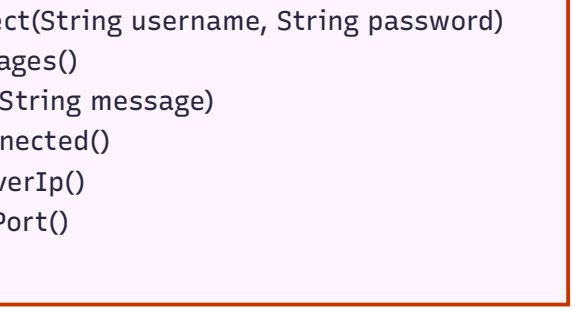

-String serverIp
-int serverPort
-Socket socket
-BufferedReader
-PrintWriter ou
-boolean isConn
-Consumer<Stri
-String lastErro
-Queue<String>
+Client(String s
+setMessageLis
+String getLast

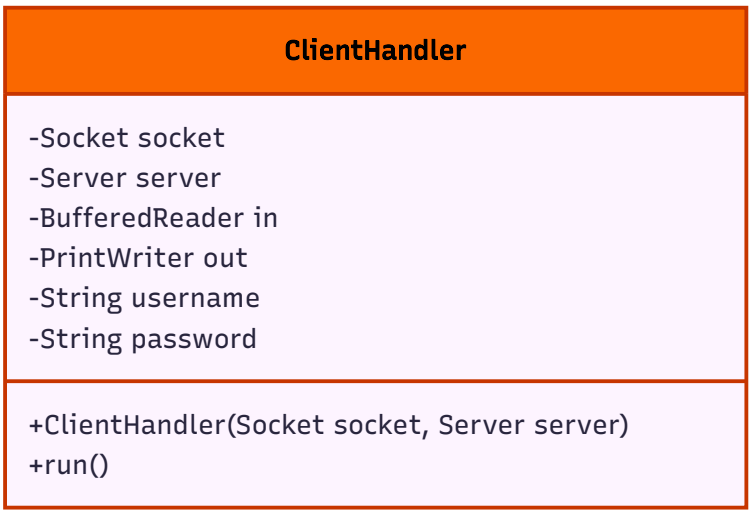

Server
-int port
-ServerSocket serverSocket
-boolean isRunning
+int defaultPort
-List<ClientHandler> clients
-Object stopLock
-boolean isStopped
+Server(int port)
+start()
+broadcast(String message, ClientHandler sender)
+sendMessageToUsername(String message, String userna
+broadcastUserList()
+removeClient(ClientHandler client)
+getClients() List<ClientHandler>
+getServerIpAddress() String
+getLocalLanIp()$ String
+getAllLanIps()$ List<String>
+getPort() int
+isRunning() boolean
+stop()
+getConnectionCode() String
Client
p
r in
ıt
nected
ing> messageListener
orMessage
> pendingQueue
serverIp, int serverPort)
stener(Consumer<String> listener)
tErrorMessage()


ame) boolean

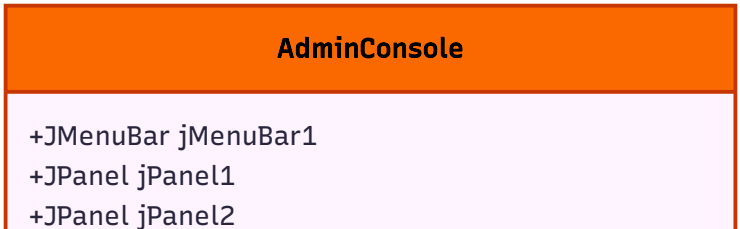

AuthService
-SecureRandom secureRandom
-long Day
-String HMAC_ALGORITHM
-String PBKDF2_ALGORITHM
-int PBKDF2_ITERATIONS
+byte[] generateSigningKey()
+String hashPassword(String password)
+boolean verifyPassword(String password, String stored)
+String generateCode(byte[] serverKey, String username, InetAddress s
+boolean verifyCode(byte[] serverKey, String code, String username, Ine
+String generateConnectionCode(String ip, int port)
+ConnectionTarget parseConnectionCode(String input)
+String[] splitCode(byte[] serverKey, String code)

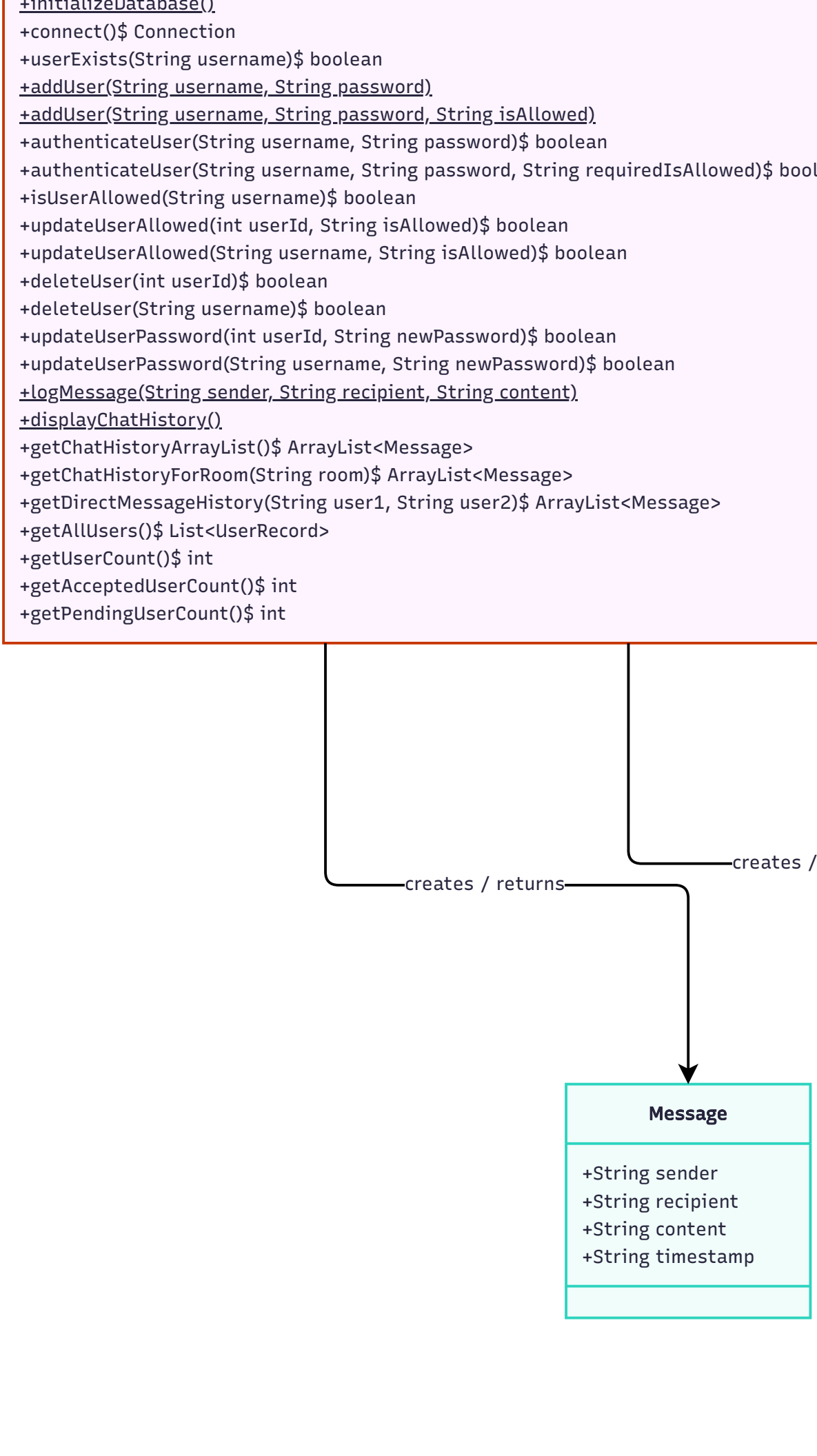

+boolean conne
-listenForMessa
+sendMessage(S
+boolean isConn
+String getServ
+int getServerP
+disconnect()
«class»
-String DB_URL
DatabaseManager


ect(String username, String password)
ages()
String message)
nected()
verIp()
Port()


ClientHandler
-Socket socket
-Server server
-BufferedReader in
-PrintWriter out
-String username
-String password
+ClientHandler(Socket socket, Server server)
+run()

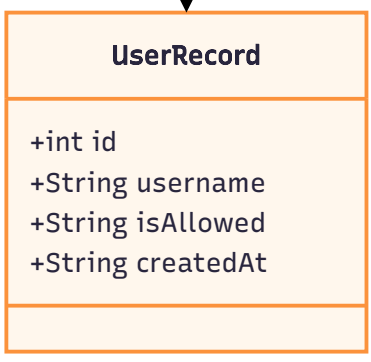



AdminConsole
+JMenuBar jMenuBar1
+JPanel jPanel1
+JPanel jPanel2


+initializeDatabase()
+connect()$ Connection
+userExists(String username)$ boolean
+addUser(String username, String password)
+addUser(String username, String password, String isAllowed)
+authenticateUser(String username, String password)$ boolean
+authenticateUser(String username, String password, String requiredIsAllowed)$ bool
+isUserAllowed(String username)$ boolean
+updateUserAllowed(int userId, String isAllowed)$ boolean
+updateUserAllowed(String username, String isAllowed)$ boolean
+deleteUser(int userId)$ boolean
+deleteUser(String username)$ boolean
+updateUserPassword(int userId, String newPassword)$ boolean
+updateUserPassword(String username, String newPassword)$ boolean
+logMessage(String sender, String recipient, String content)
+displayChatHistory()
+getChatHistoryArrayList()$ ArrayList<Message>
+getChatHistoryForRoom(String room)$ ArrayList<Message>
+getDirectMessageHistory(String user1, String user2)$ ArrayList<Message>
+getAllUsers()$ List<UserRecord>
+getUserCount()$ int
+getAcceptedUserCount()$ int
+getPendingUserCount()$ int
creates /
creates / returns
Message
+String sender
+String recipient
+String content
+String timestamp

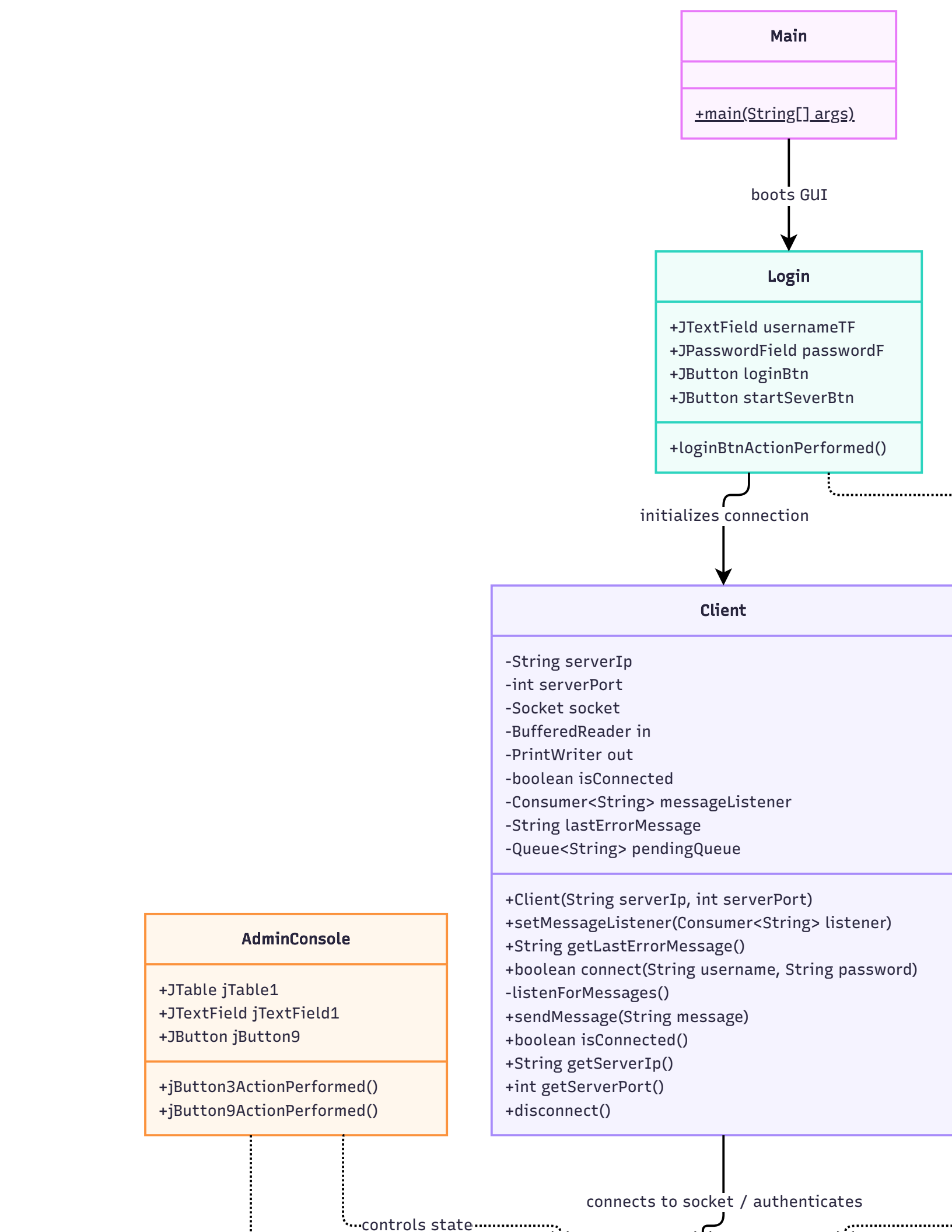
lean

/ returns

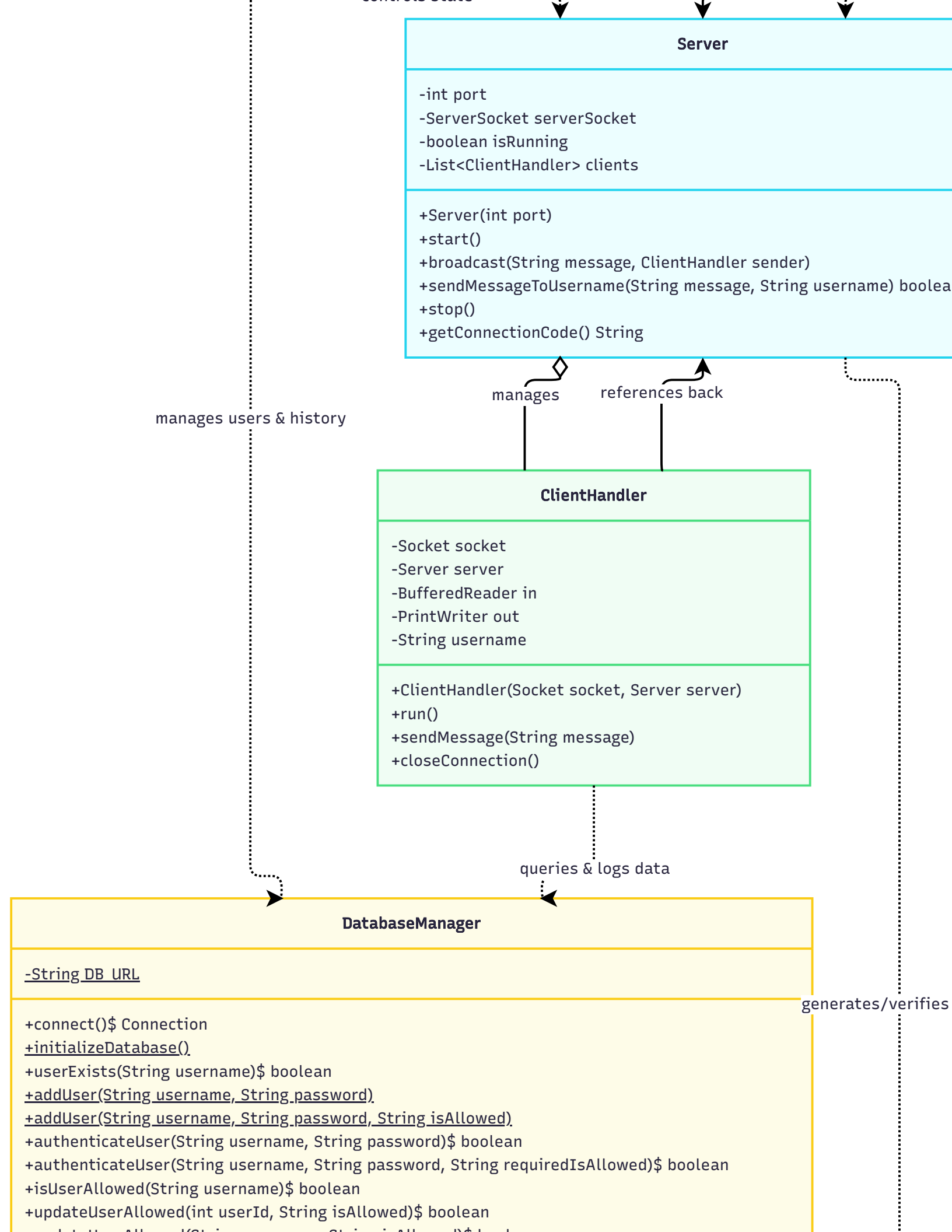


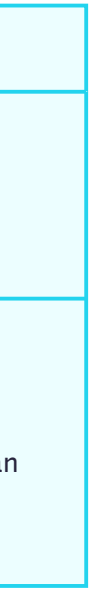
+JScrollPane jScrollPane1
+JTable jTable1
+JTextField jTextField1
+JButton jButton1
+JButton jButton2
+JButton jButton3
+JButton jButton4
+JButton jButton5
+JButton jButton6
+JButton jButton7
+JButton jButton8
+JButton jButton9

+initComponents()
+jButton3ActionPerformed(ActionEvent evt)
+jButton9ActionPerformed(ActionEvent evt)

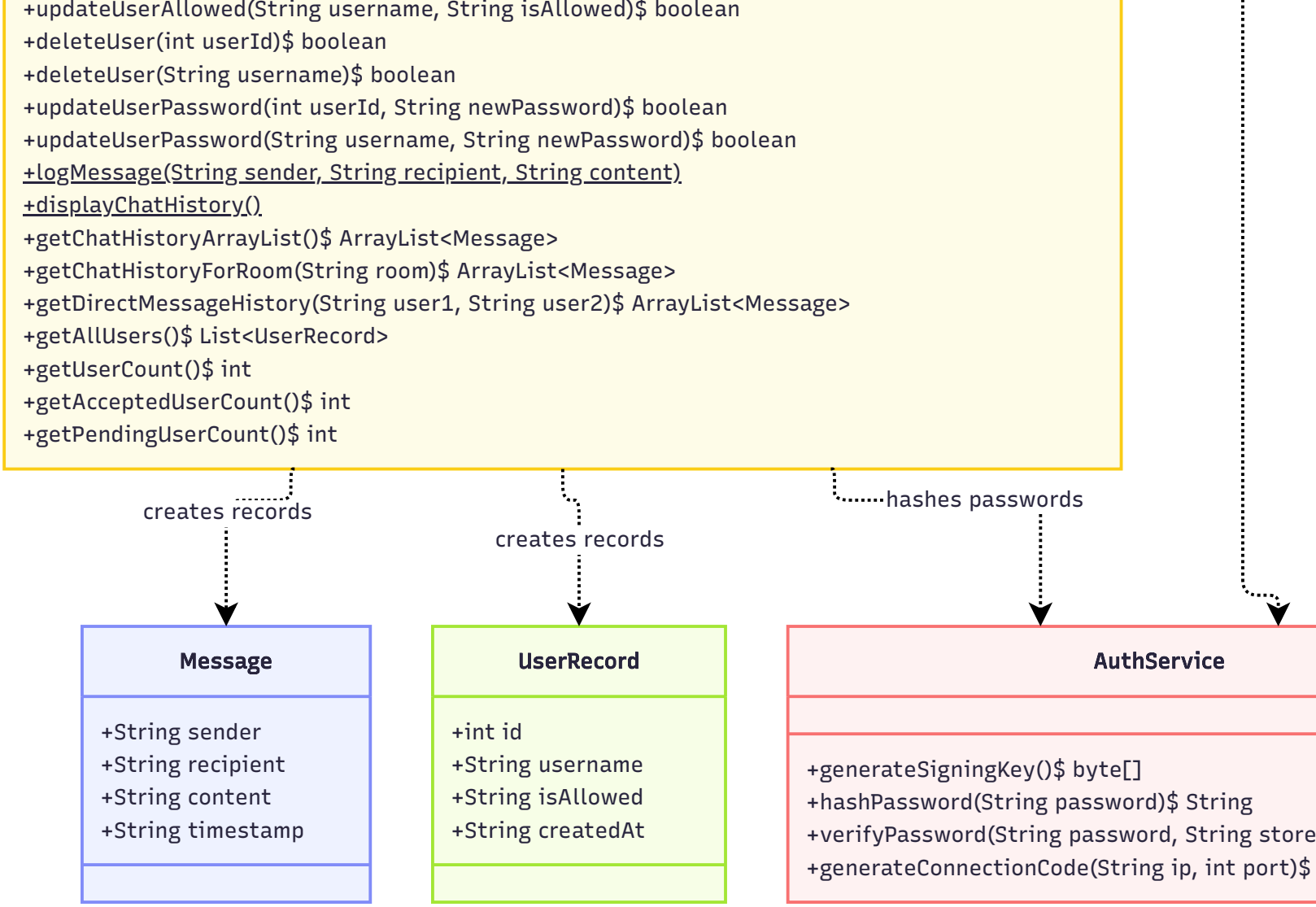

Main
+main(String[] args)
boots GUI
Login
+JTextField usernameTF
+JPasswordField passwordF
+JButton loginBtn
+JButton startSeverBtn
+loginBtnActionPerformed()
initializes connection
Client
-String serverIp
-int serverPort
-Socket socket
-BufferedReader in
-PrintWriter out
-boolean isConnected
-Consumer<String> messageListener
-String lastErrorMessage
-Queue<String> pendingQueue
+Client(String serverIp, int serverPort)
+setMessageListener(Consumer<String> listener)
+String getLastErrorMessage()
+boolean connect(String username, String password)
-listenForMessages()
+sendMessage(String message)
+boolean isConnected()
+String getServerIp()
+int getServerPort()
+disconnect()
AdminConsole
+JTable jTable1
+JTextField jTextField1
+JButton jButton9
+jButton3ActionPerformed()
+jButton9ActionPerformed()
connects to socket / authenticates
controls state

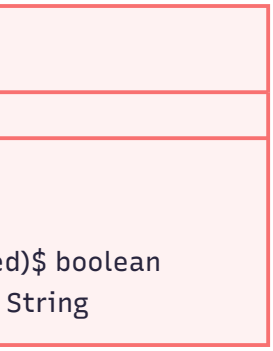

authenticates via socket


Server
-int port
-ServerSocket serverSocket
-boolean isRunning
-List<ClientHandler> clients
+Server(int port)
+start()
+broadcast(String message, ClientHandler sender)
+sendMessageToUsername(String message, String username) boolea
+stop()
+getConnectionCode() String
manages
references back
manages users & history
ClientHandler
-Socket socket
-Server server
-BufferedReader in
-PrintWriter out
-String username
+ClientHandler(Socket socket, Server server)
+run()
+sendMessage(String message)
+closeConnection()
queries & logs data
DatabaseManager
-String DB_URL
generates/verifies
+connect()$ Connection
+initializeDatabase()
+userExists(String username)$ boolean
+addUser(String username, String password)
+addUser(String username, String password, String isAllowed)
+authenticateUser(String username, String password)$ boolean
+authenticateUser(String username, String password, String requiredIsAllowed)$ boolean
+isUserAllowed(String username)$ boolean
+updateUserAllowed(int userId, String isAllowed)$ boolean



codes


+updateUserAllowed(String username, String isAllowed)$ boolean
+deleteUser(int userId)$ boolean
+deleteUser(String username)$ boolean
+updateUserPassword(int userId, String newPassword)$ boolean
+updateUserPassword(String username, String newPassword)$ boolean
+logMessage(String sender, String recipient, String content)
+displayChatHistory()
+getChatHistoryArrayList()$ ArrayList<Message>
+getChatHistoryForRoom(String room)$ ArrayList<Message>
+getDirectMessageHistory(String user1, String user2)$ ArrayList<Message>
+getAllUsers()$ List<UserRecord>
+getUserCount()$ int
+getAcceptedUserCount()$ int
+getPendingUserCount()$ int
creates records
creates records
hashes passwords
Message
+String sender
+String recipient
+String content
+String timestamp
UserRecord
+int id
+String username
+String isAllowed
+String createdAt
AuthService
+generateSigningKey()$ byte[]
+hashPassword(String password)$ String
+verifyPassword(String password, String store
+generateConnectionCode(String ip, int port)$


ed)$ boolean
String